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Hematological and biochemical responses to three doses of caffeine during the CrossFit® workout: a randomized, double-blind, crossover trial

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ABSTRACT

Background: This study investigated the effects of three doses of caffeine (CAF) on hematological and biochemical responses during the CrossFit® workout, a protocol known to induce significant physiological fluctuations.

Methods: In a randomized, double-blind, placebo (*PLA*)-controlled crossover trial, 26 moderately trained CrossFit® athletes received acute doses of 3, 6, or 9 $mg_{CAF}/kg_{body\ mass\ (BM)}$, or a PLA, 70 min before a 17-min CrossFit® workout.

Results: While exercise was the predominant driver of hematological changes, with no CAF-specific modulation of blood cell indices, significant treatment × time point interactions were observed solely for glucose concentration. Specifically, *MEDIUM* (6 mg_{CAF}/kg_{BM}) and *HIGH* (9 mg_{CAF}/kg_{BM}) CAF doses produced significantly greater post-exercise glucose elevations compared to baseline and *PLA* visits. Phosphates concentrations increased over time, particularly post-exercise, with no treatment effect retained after excluding baseline visit in the sensitivity analysis. Calcium level dynamics were influenced by temporal factors rather than CAF, and magnesium concentration remained stable.

Conclusions: These findings suggest that acute CAF ingestion selectively influences post-exercise glucose responses during the CrossFit® workout, whereas hematological variables and most biochemical markers remain unaffected. Significant post-exercise glucose elevations were observed following the 6 and 9 mg_{CAF}/kg_{BM} conditions compared with baseline and *PLA visits*, while serum mineral concentrations remained largely stable across treatments.

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1. Introduction

Caffeine (CAF) is one of the most widely consumed dietary bioactive methylxanthines. While its performance-enhancing effects have been extensively documented across endurance, strength, and high-intensity exercise domains [1,2], recent interest has expanded toward its role in modulating systemic physiological responses to acute metabolic stress. Its primary mechanisms include competitive antagonism of adenosine receptors, resulting in reduced perception of effort, increased alertness and improved cognitive processing during exercise [1,3]. Additional physiological actions such as enhanced excitation-contraction coupling, increased sarcoplasmic calcium (Ca) ions release and altered substrate utilization may also contribute to improved exercise performance [4]. Although acute ingestion of 3–6 $mg_{CAF}/kg_{body\ mass\ (BM)}$ is typically recommended for ergogenic effects, substantial inter-individual variability has been observed, influenced by genetics, habitual CAF intake and training status [1,5,6]. Accordingly, three CAF doses representing the lower (3 mg/kg_{BM}), commonly recommended (6 mg/kg_{BM}), and upper (9 mg/kg_{BM}) range of acute supplementation used in sports nutrition research were included. While these doses have been extensively investigated in relation to exercise performance,

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